

Strong disorder for 2D directed polymers and stochastic heat flow

Francesco Caravenna

University of Milano-Bicocca

Localization, Delocalization, and Diffusion in Disordered Systems

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Collaborators



Quentin Berger
(Sorbonne Paris Nord)



Nicola Turchi
(Milano-Bicocca)

Collaborators

Rongfeng Sun (NUS)

and

Nikos Zygouras (Warwick)

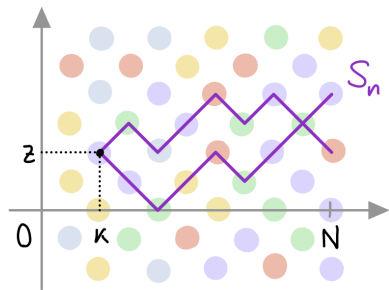


Outline

1. Directed Polymer and Stochastic Heat Flow
2. Supercritical Directed Polymer
3. Supercritical Stochastic Heat Flow
4. Sketch of the proof
5. Conclusions

Directed polymer in random environment

- ▶ $S = (S_n)_{n \geq 0}$ simple random walk on $\mathbb{Z}^d$
- ▶ Independent Gaussians $\omega(n, x) \sim \mathcal{N}(0, 1)$
- ▶ $H(S, \omega) := \sum_{n=k+1}^N \omega(n, S_n) \sim \mathcal{N}(0, N-k)$



Partition Functions

$$(k \in \mathbb{N}, z \in \mathbb{Z}^d)$$

$$Z_{N,\beta}^{\omega}(k, z) = \mathbb{E} \left[e^{\beta H(S, \omega) - \frac{1}{2} \beta^2 (N-k)} \middle| S_k = z \right]$$

$$\mathbb{E}[Z_{N,\beta}^{\omega}] = 1$$

Partition functions and SHE

Diffusive rescaling (+ time rev.)

$$u_N(t, x) := Z_{N, \beta}(N(1-t), \sqrt{N}x)$$

Discretised Stochastic Heat Equation (SHE)

$$\partial_t u_N(t, x) = \Delta_x u_N(t, x) + \beta N^{\frac{2-d}{4}} N^{\frac{2-d}{4}} \underbrace{\xi_N(t, x)}_{\text{approx. of white noise on scale } \varepsilon = 1/\sqrt{N}} u_N(t, x) \quad (\text{disc-SHE})$$

approx. of white noise on scale $\varepsilon = 1/\sqrt{N}$

Critical dimension $d = 2$:

approximation to ill-defined 2D SHE

Critical 2D Stochastic Heat Flow

[C.S.Z. 23]

$$\lim_{N \rightarrow \infty} \int u_N(t, x) dx = \mathcal{Z}_t^\vartheta(dx)$$

renormalizing $\beta = \beta_N^{\text{crit}} \rightarrow 0$

(Video)

Stochastic Heat Equation: subcritical

(by Q. Berger)

(Video)

(Video)

(Video)

From directed polymer to SHF

Averaged partition functions on balls

$$Z_{N,\beta}(f) := \sum_{z \in \mathbb{Z}^2} f(z) Z_{N,\beta}(k=0, z) \quad f(\cdot) = B_R(\cdot) = \frac{1}{\pi R^2} \mathbb{1}_{|\cdot| \leq R}$$

► Diffusive scale $R \propto \sqrt{N}$

► Critical regime $(\beta_N^{\text{crit}})^2 = \frac{\pi}{\log N} \left(1 + \frac{\vartheta + o(1)}{\log N} \right)$ for fixed $\vartheta \in \mathbb{R}$

Convergence to Stochastic Heat Flow

$$Z_{N,\beta_N^{\text{crit}}}(B_{\sqrt{N}}) \xrightarrow[N \rightarrow \infty]{d} \mathcal{Z}_t^{\vartheta}(B_1)$$

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Beyond the critical regime

Critical regime $\beta = \beta_N^{\text{crit}}$: rewrite as $\beta^2 = \frac{\pi}{\log N} \left(1 + -\frac{\vartheta + o(1)}{\log N}\right)^{-1}$

$$\boxed{\beta^2 = \frac{\pi}{\log N - \vartheta}}$$

One-to-one correspondence (given N)

$$(0, 1) \ni \beta \leftrightarrow \vartheta \in (-\infty, \log N - \pi)$$

$$\left\{ \begin{array}{lll} \text{subcritical regime} & \beta \ll \beta_N^{\text{crit}} & \vartheta \rightarrow -\infty \\ \text{critical regime} & \beta = \beta_N^{\text{crit}} & \vartheta = O(1) \\ \text{supercritical regime} & \beta \gg \beta_N^{\text{crit}} & \vartheta \rightarrow +\infty \end{array} \right.$$

Supercritical directed polymer

We prove that $Z_{N,\beta}(B_{\sqrt{N}}) \rightarrow 0$ in the **entire supercritical regime**

$$\beta \text{ “}\gg\text{” } \beta_N^{\text{crit}} \quad \text{i.e.} \quad \vartheta \rightarrow \infty$$

Quantitative bounds via $\begin{cases} \text{fractional moments} & \mathbb{E}[Z_{N,\beta}(B_{\sqrt{N}})^\gamma] \text{ for } \gamma \in (0,1) \\ \text{truncated mean} & \mathbb{E}[Z_{N,\beta}(B_{\sqrt{N}}) \wedge 1] \end{cases}$

Main Theorem

[Berger C. Turchi 25+]

$$c \exp(-C e^\vartheta) \leq \mathbb{E}[Z_{N,\beta}(B_{\sqrt{N}}) \wedge 1] \leq C \exp(-c e^\vartheta) \quad \forall N, \beta$$

LB: 2nd moment method

UB: coarse-graining + change of measure

Tail bounds and free energy

Right tail bounds

for $\varepsilon \in (0, 1)$

$$c_\varepsilon \exp(-C e^{\vartheta}) \leq \mathbb{P}(Z_{N,\beta}(B_{\sqrt{N}}) \geq \varepsilon) \leq C_\varepsilon \exp(-c e^{\vartheta})$$

Free energy (point-to-plane)

[Berger Lacoin 17]

$$F(\beta) := \lim_{N \rightarrow \infty} \frac{1}{N} \log Z_{N,\beta}(0) = \exp\left(-\frac{\pi + o(1)}{\beta^2}\right) \quad \text{as } \beta \downarrow 0$$

Free energy asymptotics

[Berger C. Turchi 25+]

$$-\frac{c}{\beta^8} \exp\left(-\frac{\pi}{\beta^2}\right) \leq F(\beta) \leq -c \exp\left(-\frac{\pi}{\beta^2}\right) \quad \left[e^{\vartheta} = N e^{-\pi/\beta^2} \right]$$

Back to discretized SHE

Solution of discretized SHE

$$\varphi \in C_c^\infty(\mathbb{R}^2)$$

$$u_{N,\beta}(t,x) = Z_{N,\beta}(N(1-t), \sqrt{N}x)$$

$$\int \varphi(x) u_{N,\beta}(t,x) dx \xrightarrow[N \rightarrow \infty]{d} \begin{cases} \int \varphi(x) dx & \text{for } \beta \ll \beta_N^{\text{crit}} \\ \int \varphi(x) \mathcal{Z}_t^\vartheta(dx) & \text{for } \beta = \beta_N^{\text{crit}} \\ 0 & \text{for } \beta \gg \beta_N^{\text{crit}} \end{cases}$$

Local extinction for $\beta \gg \beta_N^{\text{crit}}$

(e.g. for fixed $\beta > 0$)

Superdiffusivity of the supercritical SHE

Mass “conserved on average”: $\int U_R(x) u_{N,\beta}(t,x) dx \xrightarrow[R \rightarrow \infty]{d} 1$ (fixed N, β, t)

Mass escape to infinity in the **supercritical regime** $\beta \gg \beta_N^{\text{crit}}$

Theorem

[Berger C. Turchi 25+]

Rate of mass escape $\exp(A t e^{\vartheta}) = \exp(A t N e^{-\pi/\beta^2})$ ($\beta^2 = \frac{\pi}{\log N - \vartheta}$)

$\exists 0 < A_- < A_+ < \infty$:

$$\int \varphi(x) u_{N,\beta}(t, e^{A t N e^{-\pi/\beta^2}} x) dx \xrightarrow[N \rightarrow \infty]{d} \begin{cases} \int \varphi(x) dx & \text{if } A > A_+ \\ 0 & \text{if } A < A_- \end{cases}$$

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Local extinction

Analogous results for the SHF $\mathcal{Z}_t^\vartheta$

- SHF vanishes for long time $t \rightarrow \infty$ [C.S.Z. 25+]

$$\forall \varphi \in C_c(\mathbb{R}^2): \quad \mathcal{Z}_t^\vartheta(\varphi) = \int_{\mathbb{R}^2} \varphi(x) \mathcal{Z}_t^\vartheta(dx) \xrightarrow[t \rightarrow \infty]{d} 0$$

- SHF vanishes for strong disorder $\vartheta \rightarrow \infty$ [Clark Tsai 25+]

$$\mathcal{Z}_t^\vartheta(\varphi) \xrightarrow[\vartheta \rightarrow \infty]{d} 0$$

We obtain quantitative bounds. For simplicity $\varphi(\cdot) = B_r(\cdot) = \frac{1}{\pi r^2} \mathbb{1}_{|\cdot| \leq r}$

Quantitative bounds

Theorem

[Berger C. Turchi 25+]

$$c \exp(-C t e^{\vartheta}) \leq \mathbb{E}[\mathcal{Z}_t^{\vartheta}(B_{\sqrt{t}}) \wedge 1] \leq C \exp(-c t e^{\vartheta}) \quad \forall t, \vartheta$$

In ongoing work, we extend this bound to B_r uniformly for $0 < r \leq \sqrt{t}$

Theorem ($\gamma \in (0, 1)$)

[Berger C. Sun Turchi Zygouras 26+]

$$c \mathbb{E}[\mathcal{Z}_t^{\vartheta}(B_r)^2]^{-C\gamma(1-\gamma)} \leq \mathbb{E}[\mathcal{Z}_t^{\vartheta}(B_r)^{\gamma}] \leq C \mathbb{E}[\mathcal{Z}_t^{\vartheta}(B_r)^2]^{-c\gamma(1-\gamma)} \quad \forall t, \vartheta, r$$

LB agrees with log-log fluctuations of $\mathcal{Z}_t^{\vartheta}(B_r)$ as $r \downarrow 0$

[Gu Tsai 26+]

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Proof of the UB: coarse-graining

$$\mathbb{E}\left[Z_{N,\beta}(B_{e^{\hat{c}}e^{\vartheta}\sqrt{N}}) \wedge 1\right] \leq C \exp(-ce^{\vartheta})$$

- Change of scale argument: reduce $B_{e^{\hat{c}}e^{\vartheta}\sqrt{N}}$ to $B_{\sqrt{N}}$
- Coarse-graining argument: $\begin{cases} \text{reduce } \vartheta \rightarrow \infty \text{ to fixed (large) } \vartheta > 0 \\ \text{replace } \exp(-ce^{\vartheta}) \text{ by any } f(\vartheta) \rightarrow 0 \end{cases}$

Key bound

$$\mathbb{E}\left[Z_{N,\beta}(B_{\sqrt{N}}) \wedge 1\right] \leq \frac{C}{\vartheta} \quad \text{for fixed } \vartheta \in \mathbb{R}$$

Proof of the UB: change of measure

Change of scale: $\mathbb{E}[Z_{N,\beta}(B_{\sqrt{N}}) \wedge 1] \leq \frac{C}{\varepsilon} \mathbb{E}[Z_{N,\beta}(B_{\sqrt{\varepsilon N}}) \wedge 1] \quad \forall \varepsilon \in (0,1)$

For small $\varepsilon > 0$ the partition function $Z_{N,\beta}(B_{\sqrt{\varepsilon N}})$ is almost **point-to-plane**

Size-biased law $\tilde{\mathbb{P}}(d\omega) := Z(\omega) \mathbb{P}(d\omega) \quad Z(\omega) := Z_{N,\beta}(B_{\sqrt{\varepsilon N}})$

Change of measure

$$\mathbb{E}[Z \wedge 1] \leq \mathbb{P}(A) + \tilde{\mathbb{P}}(A^c) \quad \text{for any event } A$$

Goal: find some event A which is **typical** for $\mathbb{P}$ and **atypical** for $\tilde{\mathbb{P}}$

Optimal choice $A = \{Z > 1\}$ (but we don't know Z)

Proof of the UB: choice of a proxy

Take $A = \{X > \frac{1}{2} \tilde{\mathbb{E}}[X]\}$ for some r.v. X with $\mathbb{E}[X] = 0$ ($X \approx Z - 1$)

By Markov and Chebychev $\mathbb{P}(A) \leq 4 \frac{\text{Var}[X]}{\tilde{\mathbb{E}}[X]^2}$ $\tilde{\mathbb{P}}(A^c) \leq 4 \frac{\tilde{\text{Var}}[X]}{\tilde{\mathbb{E}}[X]^2}$

We take X as a **manageable proxy** of Z restricting its **chaos expansion** to

$$I = \{(n_1, x_1), \dots, (n_k, x_k)\} \quad \text{with} \quad \begin{cases} \text{width}(I) = n_k - n_1 \leq \varepsilon N, \\ |I| = k \leq \log(\varepsilon N) \end{cases}$$

We finally estimate $\text{Var}[X]$, $\tilde{\mathbb{E}}[X]$ (2nd moment) and $\tilde{\text{Var}}[X]$ (3rd moment)

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Conclusions

Quantitative bounds for **local extinction** of SHF and directed polymers
large time and/or strong disorder

Mass escape to infinity on the **spatial scale** $\exp(ct e^{\vartheta}) = \exp(ct N e^{-\pi/\beta^2})$

Application: **discretized SHE** in the supercritical regime (including **fixed** $\beta > 0$)
We expect analogous result for the **mollified SHE** (in progress)

Robust proof based on **coarse-graining** + **change of scale** + **change of measure**

Thanks